LAB SHEET 3

Experiment Name - > Statistical Analysis and Hypothesis Testing using Python

Steps

1.
Select a real-world dataset such as the IBM HR Employee Attrition Dataset, Titanic Survival Dataset, or House Price Prediction Dataset.

2.
Import the required Python libraries including Pandas, NumPy, SciPy, Matplotlib, and Statsmodels.

3.
Load the dataset and calculate descriptive statistics such as mean, median, mode, variance, and standard deviation for relevant numerical features.

4.
Analyze relationships between variables by calculating the Pearson correlation coefficient and visualizing the correlation matrix.

5.
Formulate a Null Hypothesis (H_0) and an Alternative Hypothesis (H_1) for a business problem (e.g., "Is there a significant difference in monthly income between employees who leave the company and those who stay?").

6.
Perform an Independent Sample t-test to compare the means of two groups and interpret the p-value.

7.
Apply One-Way ANOVA to determine whether there is a statistically significant difference among multiple groups (e.g., employee satisfaction across different job roles).

8.
Build a simple Linear Regression model to analyze the relationship between two numerical variables, such as Years of Experience and Monthly Income or House Area and House Price.

9.
Interpret the statistical outputs, including p-values, confidence intervals, regression coefficients, and R^2 score.

10.
Summarize the findings and discuss how statistical analysis can support business decisions and predictive analytics.

Solutions->

STEP 1 : Select and Load the Dataset

```
import pandas as pd
url = "https://raw.githubusercontent.com/nelson-wu/employee-attrition-ml/master/WA_Fn-UseC_-HR-Employee-Attrition.csv"
df = pd.read_csv(url)
print("Dataset loaded successfully!")
print("Dataset Shape:", df.shape)
display(df.head())
print("Number of rows:", df.shape[0])
print("Number of columns:", df.shape[1])
print("\nColumn Names:")
print(df.columns.tolist())
print("\nMissing Values:")
print(df.isnull().sum().sum())
```

```
Dataset loaded successfully!
Dataset Shape: (1470, 35)
```

	Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHome	Education	EducationField	EmployeeCount	EmployeeNumber	...	RelationshipSatisfaction	StandardHours	StockOptionLevel	TotalWorkingYears	Traini
0	41	Yes	Travel_Rarely	1102	Sales	1	2	Life Sciences	1	1	...	1	80	0	8	
1	49	No	Travel_Frequently	279	Research & Development	8	1	Life Sciences	1	2	...	4	80	1	10	
2	37	Yes	Travel_Rarely	1373	Research & Development	2	2	Other	1	4	...	2	80	0	7	
3	33	No	Travel_Frequently	1392	Research & Development	3	4	Life Sciences	1	5	...	3	80	0	8	
4	27	No	Travel_Rarely	591	Research & Development	2	1	Medical	1	7	...	4	80	1	6	

5 rows × 35 columns

```
Number of rows: 1470
Number of columns: 35

Column Names:
['Age', 'Attrition', 'BusinessTravel', 'DailyRate', 'Department', 'DistanceFromHome', 'Education', 'EducationField', 'EmployeeCount', 'EmployeeNumber', 'EnvironmentSatisfaction', 'Gender', 'HourlyRate', 'JobInvolvement', '

Missing Values:
```

STEP 2 : Import Required Python Libraries

```
import pandas as pd
import numpy as np
import scipy.stats as stats
import matplotlib.pyplot as plt
import statsmodels.api as sm
import statsmodels.formula.api as smf
pd.set_option('display.max_columns', None)
print("All required libraries imported successfully.")
```

```
All required libraries imported successfully.
```

STEP 3 : Descriptive Statistics

```
numerical_features = [
    'Age',
    'MonthlyIncome',
    'YearsAtCompany',
    'JobSatisfaction'
]
descriptive_statistics = pd.DataFrame({
    'Mean': df[numerical_features].mean(),
    'Median': df[numerical_features].median(),
    'Mode': df[numerical_features].mode().iloc[0],
    'Variance': df[numerical_features].var(),
    'Standard Deviation': df[numerical_features].std()
})
print("Descriptive Statistics:")
display(descriptive_statistics)
```

Descriptive Statistics:

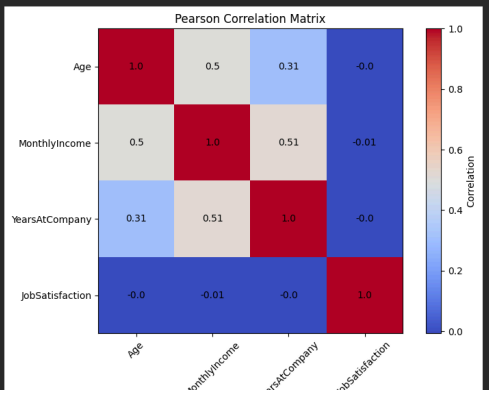
	Mean	Median	Mode	Variance	Standard Deviation
Age	36.923810	36.0	35	8.345505e+01	9.135373
MonthlyIncome	6502.931293	4919.0	2342	2.216486e+07	4707.956783
YearsAtCompany	7.008163	5.0	5	3.753431e+01	6.126525
JobSatisfaction	2.728571	3.0	4	1.216270e+00	1.102846

STEP 4 : Pearson Correlation and Correlation Matrix

```
correlation_matrix = df[numerical_features].corr(method='pearson')
print("Pearson Correlation Matrix:")
display(correlation_matrix)
plt.figure(figsize=(8, 6))
plt.imshow(correlation_matrix, cmap='coolwarm', interpolation='nearest')
plt.colorbar(label='Correlation')
plt.xticks(
    range(len(correlation_matrix.columns)),
    correlation_matrix.columns,
    rotation=45
)
plt.yticks(
    range(len(correlation_matrix.columns)),
    correlation_matrix.columns
)
for i in range(len(correlation_matrix.columns)):
    for j in range(len(correlation_matrix.columns)):
        plt.text(
            j,
            i,
            round(correlation_matrix.iloc[i, j], 2),
            ha='center',
            va='center'
        )
plt.title("Pearson Correlation Matrix")
plt.tight_layout()
plt.show()
correlation, p_value = stats.pearsonr(
    df['YearsAtCompany'],
    df['MonthlyIncome']
)
print("Pearson Correlation:", correlation)
print("P-value:", p_value)
```

```
Pearson Correlation Matrix:
```

	Age	MonthlyIncome	YearsAtCompany	JobSatisfaction
Age	1.000000	0.497855	0.311309	-0.004892
MonthlyIncome	0.497855	1.000000	0.514285	-0.007157
YearsAtCompany	0.311309	0.514285	1.000000	-0.003803
JobSatisfaction	-0.004892	-0.007157	-0.003803	1.000000



STEP 5 and 6 : Formulate Null and Alternative Hypotheses

```
[ ]
income_stayed = df[df['Attrition'] == 'No']['MonthlyIncome']
income_left = df[df['Attrition'] == 'Yes']['MonthlyIncome']
print("Number of employees who stayed:", len(income_stayed))
print("Number of employees who left:", len(income_left))
print("Average Monthly Income - Employees who stayed:",
      income_stayed.mean())
print("Average Monthly Income - Employees who left:",
      income_left.mean())
t_statistic, p_value = stats.ttest_ind(
    income_stayed,
    income_left,
    equal_var=False
)
print("T-statistic:", t_statistic)
print("P-value:", p_value)
alpha = 0.05
if p_value < alpha:
    print("Reject the Null Hypothesis (H0).")
    print("There is a statistically significant difference in monthly income between the two groups.")
else:
    print("Fail to Reject the Null Hypothesis (H0).")
    print("There is no statistically significant difference in monthly income between the two groups.")
plt.figure(figsize=(8, 6))
df.boxplot(
    column='MonthlyIncome',
    by='Attrition'
)
plt.title("Monthly Income by Employee Attrition")
plt.suptitle("")
plt.xlabel("Attrition")
plt.ylabel("Monthly Income")
plt.show()
```

STEP 7 : One-Way ANOVA

```
[ ]
print("Different Job Roles:")
print(df['JobRole'].unique())
groups = []
job_roles = df['JobRole'].unique()
for role in job_roles:
    group = df[df['JobRole'] == role]['JobSatisfaction']
    groups.append(group)
print("Number of groups:", len(groups))
f_statistic, anova_p_value = stats.f_oneway(*groups)
print("F-statistic:", f_statistic)
print("P-value:", anova_p_value)
alpha = 0.05
if anova_p_value < alpha:
    print("Reject the Null Hypothesis (H0).")
    print("There is a statistically significant difference in job satisfaction among job roles.")
else:
    print("Fail to Reject the Null Hypothesis (H0).")
    print("There is no statistically significant difference in job satisfaction among job roles.")
model_anova = smf.ols(
    'JobSatisfaction ~ C(JobRole)',
    data=df
).fit()
anova_table = sm.stats.anova_lm(model_anova, typ=2)
print("ANOVA Table:")
display(anova_table)
```

```
Different Job Roles:
['Sales Executive' 'Research Scientist' 'Laboratory Technician'
 'Manufacturing Director' 'Healthcare Representative' 'Manager'
 'Sales Representative' 'Research Director' 'Human Resources']
Number of groups: 9
F-statistic: 0.3652125646864016
P-value: 0.9389893610579624
Fail to Reject the Null Hypothesis (H0).
There is no statistically significant difference in job satisfaction among job roles.
ANOVA Table:
```

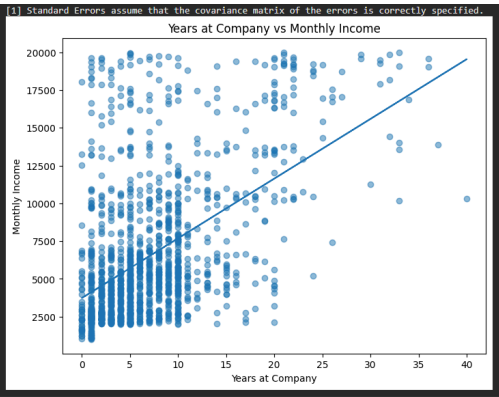
	sum_sq	df	F	PR(>F)
C(JobRole)	3.565903	8.0	0.365213	0.938989
Residual	1783.134097	1461.0	NaN	NaN

STEP 8 : Simple Linear Regression

```
X = df['YearsAtCompany']
Y = df['MonthlyIncome']
X_with_constant = sm.add_constant(X)
regression_model = sm.OLS(Y, X_with_constant).fit()
print(regression_model.summary())
plt.figure(figsize=(8, 6))
plt.scatter(
    df['YearsAtCompany'],
    df['MonthlyIncome'],
    alpha=0.5
)
plt.plot(
    df['YearsAtCompany'],
    regression_model.predict(X_with_constant)
)
plt.xlabel("Years at Company")
plt.ylabel("Monthly Income")
plt.title("Years at Company vs Monthly Income")
plt.show()
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:          MonthlyIncome   R-squared:                       0.264
Model:                            OLS   Adj. R-squared:                  0.264
Method:                 Least Squares   F-statistic:                     527.9
Date:                Mon, 07 Sep 2026   Prob (F-statistic):          4.82e-100
Time:                        08:38:59   Log-Likelihood:                -14291.
No. Observations:                1470   AIC:                         2.859e+04
Df Residuals:                    1468   BIC:                         2.860e+04
Df Model:                           1
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const          3733.2731    160.091     23.320      0.000    3419.242    4047.305
YearsAtCompany   395.2046     17.201     22.976      0.000     361.464     428.945
==============================================================================
Omnibus:                      387.315   Durbin-Watson:                   2.034
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              875.316
Skew:                           1.462   Prob(JB):                     8.46e-191
Kurtosis:                       5.395   Cond. No.                         14.2
==============================================================================

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```



STEP 9 : Interpret Statistical Outputs

```
print("Regression Statistics")
print("---------------------")
print("R-squared:",
      round(regression_model.rsquared, 4))
print("Adjusted R-squared:",
      round(regression_model.rsquared_adj, 4))
print("F-statistic:",
      round(regression_model.fvalue, 4))
print("F-test p-value:",
      regression_model.f_pvalue)
print("YearsAtCompany coefficient:",
      regression_model.params['YearsAtCompany'])
print("YearsAtCompany p-value:",
      regression_model.pvalues['YearsAtCompany'])
```

STEP 10 : Final Summary and Business Findings

```
print("=" * 60)
print("STATISTICAL ANALYSIS SUMMARY")
print("=" * 60)
print("\n1. INDEPENDENT SAMPLE T-TEST")
print("---------------------------------")
income_stayed = df[df['Attrition'] == 'No']['MonthlyIncome']
income_left = df[df['Attrition'] == 'Yes']['MonthlyIncome']
mean_stayed = income_stayed.mean()
mean_left = income_left.mean()
t_statistic, t_p_value = stats.ttest_ind(
    income_stayed,
    income_left,
    equal_var=False
)
print("Mean income - Stayed:", round(mean_stayed, 2))
print("Mean income - Left:", round(mean_left, 2))
print("T-statistic:", round(t_statistic, 4))
print("P-value:", round(t_p_value, 6))
if t_p_value < 0.05:
    print("Decision: Reject H0")
    print("There is a statistically significant difference")
    print("in monthly income between employees who stayed and left.")
else:
    print("Decision: Fail to Reject H0")
    print("There is no statistically significant difference")
    print("in monthly income between employees who stayed and left.")
print("\n2. ONE-WAY ANOVA")
print("-------------------------------")
anova_model = smf.ols(
    'JobSatisfaction ~ C(JobRole)',
    data=df
).fit()
anova_table = sm.stats.anova_lm(
    anova_model,
    typ=2
)
display(anova_table)
anova_p_value = anova_table['PR(>F)'].iloc[0]
print("ANOVA P-value:", round(anova_p_value, 6))
if anova_p_value < 0.05:
    print("Decision: Reject H0")
    print("Job satisfaction differs significantly among job roles.")
else:
    print("Decision: Fail to Reject H0")
```

```
        print("No statistically significant difference was found")
        print("in job satisfaction among job roles.")
    print("\n3. LINEAR REGRESSION")
    print("--------------------------------")
    X = df['YearsAtCompany']
    Y = df['MonthlyIncome']
    X_with_constant = sm.add_constant(X)
    regression_model = sm.OLS(
        Y,
        X_with_constant
    ).fit()
    coefficient = regression_model.params['YearsAtCompany']
    regression_p_value = regression_model.pvalues['YearsAtCompany']
    r_squared = regression_model.rsquared
    print("Intercept:", round(regression_model.params['const'], 4))
    print("YearsAtCompany Coefficient:", round(coefficient, 4))
    print("P-value:", round(regression_p_value, 6))
    print("R-squared:", round(r_squared, 4))
    print("Adjusted R-squared:",
          round(regression_model.rsquared_adj, 4))
    print("\n95% Confidence Intervals:")
    display(regression_model.conf_int())
    if regression_p_value < 0.05:
        print("Conclusion: YearsAtCompany has a statistically")
        print("significant relationship with MonthlyIncome.")
    else:
        print("Conclusion: YearsAtCompany does not have a")
        print("statistically significant relationship with MonthlyIncome.")
```

```
============================================================
STATISTICAL ANALYSIS SUMMARY
============================================================

1. INDEPENDENT SAMPLE T-TEST
--------------------------------
Mean Income - Stayed: 6832.74
Mean Income - Left: 4787.09
T-statistic: 7.4826
P-value: 0.0
Decision: Reject H0
There is a statistically significant difference
in monthly income between employees who stayed and left.
```

```
2. ONE-WAY ANOVA
--------------------------------
```

	sum_sq	df	F	PR(>F)
C(JobRole)	3.565903	8.0	0.365213	0.938989
Residual	1783.134097	1461.0	NaN	NaN

```
ANOVA P-value: 0.938989
Decision: Fail to Reject H0
No statistically significant difference was found
in job satisfaction among job roles.

3. LINEAR REGRESSION
--------------------------------
Intercept: 3733.2731
YearsAtCompany Coefficient: 395.2046
P-value: 0.0
R-squared: 0.2645
Adjusted R-squared: 0.264

95% Confidence Intervals:
```

	0	1
const	3419.241742	4047.304554
YearsAtCompany	361.463709	428.945429

```
Conclusion: YearsAtCompany has a statistically
significant relationship with MonthlyIncome.

============================================================
OVERALL CONCLUSION
============================================================

The IBM HR Employee Attrition dataset was analyzed using
descriptive statistics, Pearson correlation, an independent
sample t-test, One-Way ANOVA, and simple linear regression.

The t-test was used to compare monthly income between
employees who stayed and employees who left.

ANOVA was used to determine whether job satisfaction differs
among different job roles.
```

Questions

1. What is the difference between descriptive statistics and inferential statistics?
2. Explain the concepts of the Null Hypothesis (H_0) and Alternative Hypothesis (H_1).
3. What is a p-value? How is it used to make statistical decisions?
4. Differentiate between a t-test and ANOVA. In which situations is each test applied?
5. What does the Pearson correlation coefficient indicate? What are its possible values?
6. Explain the significance of R^2 (Coefficient of Determination) in Linear Regression.
7. Why is statistical analysis important before applying machine learning algorithms?
8. What assumptions should be satisfied before performing a t-test or ANOVA?

9. How can regression analysis help organizations in forecasting and decision-making?
10. Give two real-world applications where hypothesis testing is commonly used in data analytics.

Solutions -> 1)

Descriptive Statistics	**Inferential Statistics**
Describes existing data	Draws conclusions about a population
Does not generalize beyond the dataset	Generalizes from sample to population
Uses mean, median, SD, etc.	Uses hypothesis tests, confidence intervals, etc.
Example: Average marks of a class	Example: Testing whether a teaching method improves marks

2) A **hypothesis** is a statement about a population that can be tested statistically.

Null Hypothesis (H_0):
It states that there is **no significant difference, relationship, or effect**.

Alternative Hypothesis (H_1):
It states that there **is a significant difference, relationship, or effect**.

Example: Suppose we want to determine whether a new teaching method improves students' marks.

- **H_0:** The new teaching method does not improve students' marks.
- **H_1:** The new teaching method improves students' marks.

After performing a statistical test, we either **reject H_0** or **fail to reject H_0**.

3) A **p-value** is the probability of obtaining the observed result, or a more extreme result, **assuming that the null hypothesis (H_0) is true**.

It is commonly compared with a significance level **$\alpha = 0.05$**.

- If **p-value < 0.05** → Reject H_0 → Result is statistically significant.
- If **p-value ≥ 0.05** → Fail to reject H_0 → Result is not statistically significant.

Example:

Suppose a statistical test gives:

p-value = 0.03

Since:

0.03 < 0.05

we reject H_0 and conclude that there is statistically significant evidence supporting H_1.

A p-value does **not** tell us the probability that H_0 itself is true.

4)

t-test	**ANOVA**
Usually compares 2 groups	Usually compares 3 or more groups
Uses t-statistic	Uses F-statistic
Example: Group A vs Group B	Example: Group A vs B vs C
Tests difference between two means	Tests whether at least one group mean differs

5) The **Pearson correlation coefficient (r)** measures the **strength and direction of a linear relationship** between two numerical variables.

Its value ranges from:

−1 ≤ r ≤ +1

- **r = +1** → Perfect positive linear correlation
- **r = −1** → Perfect negative linear correlation

- **r = 0** → No linear correlation
- **r > 0** → Positive relationship
- **r < 0** → Negative relationship

For example, if:

r = 0.85

there is a strong positive linear relationship between the variables.

6) **R^2 (Coefficient of Determination)** measures how much of the variation in the dependent variable is explained by the regression model.

Its value generally ranges from **0 to 1**.

For example:

$R^2 = 0.80$

means that approximately **80% of the variation in the dependent variable is explained by the predictors in the model**, while the remaining 20% is due to other factors and unexplained variation.

Example:
If a model predicts house prices using area, number of rooms, and location and has:

$R^2 = 0.75$

then the model explains approximately **75% of the observed variation in house prices**.

A higher R^2 generally indicates a better fit, although **a high R^2 alone does not guarantee that the model is appropriate or that the relationship is causal**.

7) Statistical analysis is important before applying machine learning because it helps us **understand and prepare the data**.

Major benefits include:

1. **Understanding data distribution** – Helps identify mean, spread, skewness, etc.
2. **Detecting outliers** – Extreme values can negatively affect some models.

3. **Finding relationships** – Correlation analysis can identify relationships between variables.
4. **Feature selection** – Irrelevant or redundant features can sometimes be removed.
5. **Detecting missing values and data-quality problems.**
6. **Checking assumptions** – Some statistical and ML techniques require specific assumptions.
7. **Understanding the target variable** – Helps determine whether regression, classification, or another approach is suitable.
8. **Avoiding misleading conclusions** – Statistical significance and practical significance can be considered before modeling.

Therefore, statistical analysis provides a better understanding of the dataset and can improve the quality of the machine learning workflow.

8) Common assumptions include:

1. **Independence**
   Observations should be independent of each other.
2. **Continuous dependent variable**
   The variable being compared should generally be numerical and continuous.
3. **Normality**
   The data, or more precisely the relevant residuals/sampling distributions, should be approximately normally distributed, especially for small samples.
4. **Homogeneity of variance**
   The groups should have approximately equal variances for the standard versions of these tests.
5. **Random sampling**
   Ideally, observations should be obtained through an appropriate random or representative sampling process.

If these assumptions are seriously violated, alternatives such as **Mann–Whitney U test** or **Kruskal–Wallis test** may be considered, depending on the situation.

9) **Regression analysis** examines the relationship between a dependent variable and one or more independent variables. It can be used to estimate or predict future values.

Organizations can use regression for:

- **Sales forecasting** – Predict future sales based on previous sales, advertising, price, etc.
- **Demand forecasting** – Estimate future demand for products.
- **Financial planning** – Predict revenue, expenses, or profits.
- **Price prediction** – Estimate property or product prices.
- **Risk analysis** – Identify factors associated with financial or operational risk.
- **Resource planning** – Estimate future staffing or inventory requirements.

Example:
A company can build a regression model:

Sales = β_0 + β_1(Advertising) + β_2(Price) + β_3(Season) + ε

The organization can then estimate expected sales under different advertising budgets and pricing strategies.

Thus, regression helps organizations make **data-driven forecasts and decisions**.

10) **Application 1: Medical Research**

A pharmaceutical company wants to determine whether a **new medicine is more effective than an existing medicine**.

- **H_0:** The new medicine has no difference in effectiveness compared with the existing medicine.
- **H_1:** The new medicine has a different effectiveness.
- A suitable statistical test is performed.
- If **p-value < 0.05**, H_0 is rejected, suggesting statistically significant evidence of a difference.

This helps researchers evaluate the effectiveness of treatments using sample data.

Application 2: Business/Marketing

An e-commerce company wants to know whether a **new website design increases the conversion rate**.

- **H_0:** The new website design does not change the conversion rate.
- **H_1:** The new website design increases the conversion rate.
- Customers can be divided into control and test groups.

- Their conversion rates are statistically compared.
- If the p-value is below the chosen significance level, the company has evidence that the new design has a statistically significant effect.

This type of **A/B testing** is widely used in data analytics for making business decisions.